

Solutions - Mid Semester Test

Course: CSIZG514/SEZG514 - Introduction to DevOps

Q1. DevOps Practices to Reduce Technical Debt [5 Marks]

Practices:

- Continuous Integration (CI)
- Continuous Testing
- Continuous Delivery/Deployment (CD)
- Infrastructure as Code (IaC)
- Automated Monitoring & Feedback

How they reduce debt:

- Automating repetitive tasks reduces manual intervention.
- Early feedback on code quality reduces rework.
- Small increments avoid long-lived branches and outdated dependencies.

Tools:

- CI/CD: Jenkins, GitHub Actions, GitLab CI, CircleCI
- Testing: JUnit, Selenium, pytest
- IaC: Terraform, Ansible, Chef, Puppet
- Monitoring: Prometheus, Grafana, ELK Stack

Q2. Git Concepts [5 Marks]

(a) Advantages of Forking Workflow:

- Full isolation of development
- Safe collaboration with open-source projects
- Maintains central repo integrity
- Encourages innovation

(b) Difference between git pull and git fetch:

- git fetch: Downloads changes from remote, no merge.
- git pull: Fetch + merge (integrates changes).

(c) Merge Conflict & Handling:

- Occurs when same file/line differs across branches.
- Handle by manual resolution, editing conflicts, then add & commit.

Q3. Component-Based Student Registration System [10 Marks]

(a) Component: Independent, reusable unit with defined interfaces.

(b) Dependency Graph:

UI → Authentication → Registration Manager → Database

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→ Payment Processor → Database

Database → Analytics

(c) Pipelining Dependency Graph:

- Stage 1: Data Collection (UI + Auth)
- Stage 2: Registration Manager
- Stage 3: Payment Processing
- Stage 4: Database Update
- Stage 5: Analytics

(d) Challenges:

- Interoperability
- Dependency management
- Customization vs reuse
- Complex testing

Q4. DevOps Culture Change [5 Marks]

Yes, DevOps changes culture by collaboration, transparency, shared responsibility.

Impact:

- Developers responsible for deployment & monitoring.
- Ops engaged early (shift-left).
- Encourages trust & continuous learning.

Suggested Practices:

- Blameless postmortems
- Continuous feedback loops
- ChatOps collaboration
- Automated testing & monitoring
- Agile + DevOps alignment

Q5. DevOps vs Agile & Tool Transformation [5 Marks]

(a) Agile vs DevOps:

- Agile: Iterative development, customer feedback.
- DevOps: Development + operations integration.
- Suggestion: Training to show Agile = what/how to build, DevOps = how to deliver.

(b) Tools & Productivity:

- Tools alone don't solve bottlenecks.
- True productivity = people + processes + tools.
- Focus on collaboration, automation strategy, and gradual adoption.